

Natural Language Processing Fundamentals

What is NLP?

Natural language processing (NLP) is a field of artificial intelligence focused on enabling computers to understand, interpret, and generate human language. It combines computational linguistics with machine learning and deep learning.

Tokenization

Tokenization is the process of splitting text into smaller units called tokens, which may be words, subwords, or characters. Tokenization is usually the first preprocessing step in any NLP pipeline.

Embeddings

Word and sentence embeddings map text into dense vectors in a continuous space, where semantic similarity corresponds to geometric proximity. Embeddings allow models to reason about meaning rather than raw characters.